

Day 1 of 75 days Ethical Hacking Course

- Introduction
- Hacking labs setup
- Network fundamentals
- How websites works
- Linux fundamental
- Computer fundamentals
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- Wireless attacks
- System pentesting
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- Introduction to mobile hacking
- Some important tools in ethical hacking
- Wireshark, burpsuite, metasploit, autopsy, snort many more
- Introduction to webhacking
- Introduction to digital forensics
- Malware analysis
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- Introduction
- Hacking labs setup

NOT FOR SALE



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Cyber security

Ethical hacking

offensive security

In short, offensive security is the process of breaking into computer systems, exploiting software bugs, and finding loopholes in applications to gain unauthorized access to them.

In a defensive cyber role, you could be investigating infected computers or devices to understand how it was hacked, tracking down cybercriminals, or monitoring infrastructure for malicious activity.

Which of the following options better represents the process where you simulate a hacker's actions to find vulnerabilities in a system?

- Offensive Security
- Defensive Security

HACK YOUR FIRST MACHINE

NOT FOR SALE

This page allows an attacker to steal money from any bank account, which is a critical risk for the bank. As an ethical hacker, you would (with permission) find vulnerabilities in their application and report them to the bank to fix before a hacker exploits them.

Transfer \$2000 from the bank account 2276, to your account (account number 8881)

<http://fakebank.com/bank-transfer>

What careers are there?

The cyber careers room goes into more depth about the different careers in cyber. However, here is a short description of a few offensive security roles:

- Penetration Tester - Responsible for testing technology products for finding exploitable security vulnerabilities.
- Red Teamer - Plays the role of an adversary, attacking an organization and providing feedback from an enemy's perspective.
- Security Engineer - Design, monitor, and maintain security controls, networks, and systems to help prevent cyberattacks.

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Defensive security



Defensive security is somewhat the opposite of offensive security, as it is concerned with two main tasks:

1. Preventing intrusions from occurring
2. Detecting intrusions when they occur and responding properly

Blue teams are part of the defensive security landscape.

Some of the tasks that are related to defensive security include:

- User cyber security awareness:
- Updating and patching systems:
- Setting up preventative security devices

Skills

- Security Operations Center (SOC)
- Threat Intelligence
- Digital Forensics and Incident Response (DFIR)
- Malware Analysis

Security Operations Center (SOC)

A *Security Operations Center (SOC)* is a team of cyber security professionals that monitors the network and its systems to detect malicious cyber security events. Some of the main areas of interest for a SOC are:

- Vulnerabilities:
- Unauthorized activity
- Network intrusions:
- .

Security operations cover various tasks to ensure protection; one such task is threat intelligence.

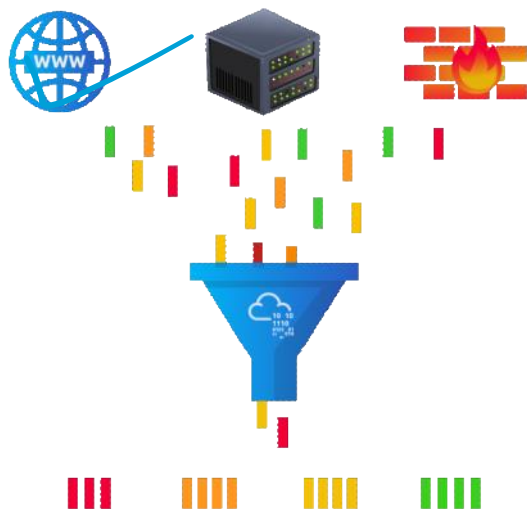
Soc analyst



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Threat Intelligence

In this context, *intelligence* refers to information you gather about actual and potential enemies. Threat intelligence aims to gather information to help the company better prepare against potential adversaries.



Intelligence needs data. Data has to be collected, processed, and analyzed. Data collection is done from local sources such as network logs and public sources such as forums. Processing of data aims to arrange them into a format suitable for analysis. The analysis phase seeks to find more information about the attackers and their motives; moreover, it aims to create a list of recommendations and actionable steps.

Learning about your adversaries allows you to know their tactics, techniques, and procedures. As a result of threat intelligence, we identify the threat actor (adversary), predict their activity, and consequently, we will be able to mitigate their attacks and prepare a response strategy.

Digital Forensics and Incident Response (DFIR)

This section is about Digital Forensics and Incident Response (DFIR), and we will cover:

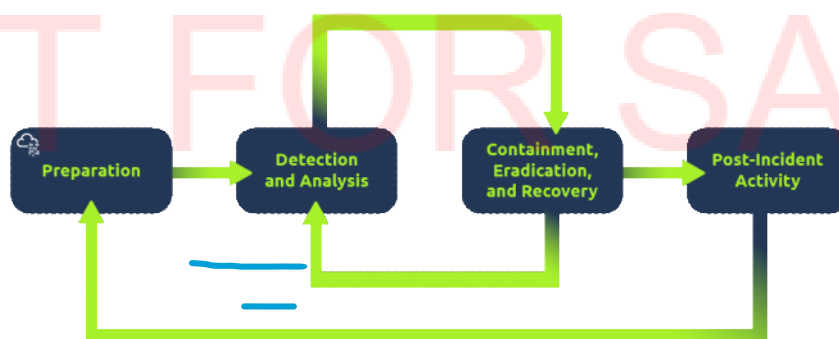
- Digital Forensics
- Incident Response
- Malware Analysis

Digital Forensics

Forensics is the application of science to investigate crimes and establish facts. With the use and spread of digital systems, such as computers and smartphones, a new branch of forensics was born to investigate related crimes: computer forensics, which later evolved into, *digital forensic*

Incident Response

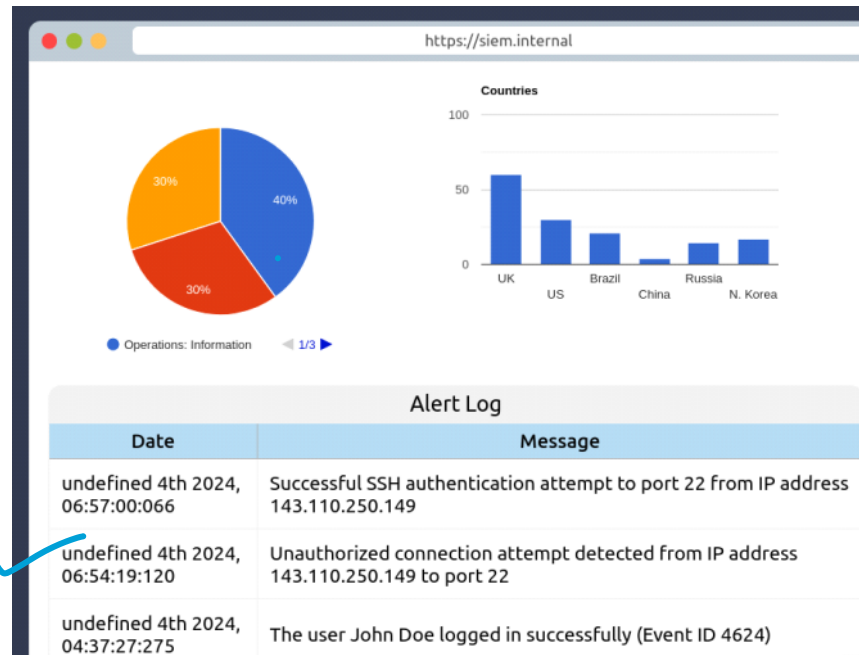
An *incident* usually refers to a data breach or cyber attack; however, in some cases, it can be something less critical, such as a misconfiguration, an intrusion attempt, or a policy violation.



Malware Analysis

Malware stands for malicious software. *Software* refers to programs, documents, and files that you can save on a disk or send over the network. Malware includes many types, such as:

- Virus is a piece of code (part of a program) that attaches itself to a program. It is designed to spread from one computer to another; moreover, it works by altering, overwriting, and deleting files once it infects a computer. The result ranges from the computer becoming slow to unusable.
- Trojan Horse is a program that shows one desirable function but hides a malicious function underneath. For example, a victim might download a video player from a shady website that gives the attacker complete control over their system.
- Ransomware is a malicious program that encrypts the user's files. Encryption makes the files unreadable without knowing the encryption password. The attacker offers the user the encryption password if the user is willing to pay a "ransom."



https://ip-scanner.thm

IP-SCANNER.THM

Check by IP Address

143.110.250.149

Instructions

There are many open-source databases out there, like AbuseIPDB, and Cisco Talos Intelligence, where you can perform a reputation and location check for the IP address. Most security analysts use these tools to aid them with alert investigations. You can also make the Internet safer by reporting the malicious IPs, for example, on AbuseIPDB.

Now that we know the IP address is malicious, we need to escalate it to a staff member!

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https://ip-scanner.thm/search



IP-SCANNER.THM

143.110.250.149 was found in our database!

Confidence of the IP being malicious is 100%

Malicious

ISP	China Mobile Communications Corporation
Domain Name	chinamobiletd.thm
Country	China
City	Zhenjiang, Jiangsu

Instructions

We shouldn't worry too much if it was a failed authentication attempt, but you probably noticed the successful authentication attempt from the malicious IP address. Let's declare a small incident event and escalate it. There is some great staff working at the company, but you wouldn't want to escalate this to the wrong person who is not in charge of your team or department.

Choose to whom you would escalate this event?

☐ Dominick Nash

Sales Executive

☐ Nadia Watson

Security Consultant

☐ Carolyn Stone

Information Security Architect

☒ Will Griffin

SOC Team Lead

Choose Staff Member

https://firewall.internal



Firewall Block List

Block List	
Date	IP Address
July 2nd 2021, 13:27:00:948	101.34.37.231
June 30th 2021, 09:12:11:857	212.38.99.12
June 23rd 2021, 23:56:28:370	213.106.84.35



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